

Aalto University

Multivariate Statistical Analysis

MCA of Individuals' Characteristics, Values, and Political Beliefs in the US

Project Report

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1 Introduction

The project explores the correspondence of political beliefs, ethnic background, religious tendencies, and racial attitudes. A particular interest of this project lies in discovering which combinations of characteristics may result in similar voting outcomes, voting for Donald Trump or not.

The dataset of this project was found from Kaggle [1]. This data is a preprocessed version of the Harvard University 2016 CCES Common Content dataset, [2], which consists of survey responses from 64,600 adults in the US. The original data contains approximately 700 variables, most of which are answers to specific questions. The preprocessed data from [1], on the other hand, contains 21 variables that have been chosen according to [4]. The particular process that has been conducted for the data can be found from [3].

According to [4], the variables selected attempt to reflect the values and beliefs of a respondent in the best possible manner. Three of the 21 variables chosen in [4] have been inferred based on the other variables of the data. As the three variables are deterministically dependent on the other data, I omit these variables. Moreover, I remove the column containing the identifiers of the participants. Table 1 lists the final 17 variables.

The data contains a vast amount of missing values. The origin of the missing values is unclear. For example, [2] reports indicate that many of the questions were only asked from a much smaller population than the population of 64600 people. Maybe this was due to the respondents voluntarily skipping the questions. In this analysis, we will dive deeper into the cause of missing values.

Variable	Description
votetrump	Whether the respondent self-reportedly voted for Donald Trump in the 2016 presidential election. 0=did not vote for Trump, 1=voted for Trump, 2=no response
agegroup	Age of the respondent in years. 1=18-24, 2=25-34, 3=35-49, 4=50-74, 5=75-95
female	Whether the respondent self-identifies as a female. 1=female, 0=not female*
collegeed	Whether the respondent has a college degree. 1=has a college degree, 0=does not have a college degree
racef	Self-identified ethnicity. White, Black, Asian, Hispanic, Native American, Middle Eastern, Other
famincr	Household income in dollars per year 1=less than 10,000, 2=10,000-19,999, 3=20,000-29,999, ..., 9=80,000-89,999, 10=100,00-119,999, 11=120,000-149,999, 12=150,000 or more**
ideo	Ideology score of the respondent. 1=very liberal, ... , 5=very conservative
pid7na	Partianship score of the respondent. 1=strong democrat, ... , 7=strong republican

bornagain	Whether the respondent self-identifies as a born-again Christian 1=identifies as a born-again Christian, 0=does not identify as a born-again Christian
religimp	The importance of religion to the respondent. 1=not at all important, ... , 4=very important
churchatd	The frequency of church attendance of the respondent. 1=never, ... , 7=more than once a week
prayerfreq	How frequently the respondent prays. 1=never, ... , 7=several times a day
angryracism	The respondent agrees that he/she is angry that racism exists. 1=strongly agrees, 2=somewhat agrees, 3=neither agrees nor disagrees, 4=somewhat disagrees , 5=strongly disagrees, 6=no response
whiteadv	The respondent agrees that he/she considers white people to have advantages over others. 1=strongly agrees, 2=somewhat agrees, 3=neither agrees nor disagrees, 4=somewhat disagrees , 5=strongly disagrees, 6=no response
fearracism	The respondent agrees that he/she fears individuals representing another ethnic group. 1=strongly disagrees, 2=somewhat disagrees, 3=neither agrees nor disagrees, 4=somewhat agrees, 5=strongly agrees, 6=no response
racerare	The respondent agrees that racism is rare in the US. 1=strongly disagrees, 2=somewhat disagrees, 3=neither agrees nor disagrees, 4=somewhat agrees, 5=strongly agrees, 6=no response

Table 1: Descriptions of variables used. **The data does disclose, whether the value 0 stands for men only or other gender identities as well. However, it may probably be safely assumed that the majority of respondents that are not characterized as females represent the gender male.* ** *The specific categories are not disclosed.*

2 Univariate analysis

We begin with univariate analysis. The data contains both strictly categorical data, that can not be ordered and categorical data that can be ordered. For example, the responses on ethnic background cannot be put in any scale, whereas responses ranging from strongly agree to strongly disagree can be thought ordinal. Figure 1 illustrates the non-ordinal variables as pie charts and ordinal variables as histograms. Moreover, Table ?? presents the modes of the strictly categorical variables, and modes, means, medians, and variances of the ordinal variables. Here, the missing values of each separate variable are simply omitted.

Statistic	votetrump	stateno	female	colleged	raceno	bornagain	agegroup	famincr	ideo	pid7na	religimp	churchatd	prayerfreq	angryracism	whiteadv	fearraces	racereare
Mod	0	5	1	0	8	0	4	4	3	1	4	1	7	1	1	1	1
Mean							3.37	6.57	3.00	3.60	2.78	2.87	4.33	1.66	2.62	2.15	2.28
Med							4	6	3	3	3	2	5	1	2	2	2
Var							0.95	9.90	1.21	4.72	1.35	2.87	5.52	0.89	2.10	1.31	1.51
Min							0.95	9.90	1.21	4.72	1.35	2.87	5.52	0.89	2.10	1.31	1.51
Var							0.95	9.90	1.21	4.72	1.35	2.87	5.52	0.89	2.10	1.31	1.51

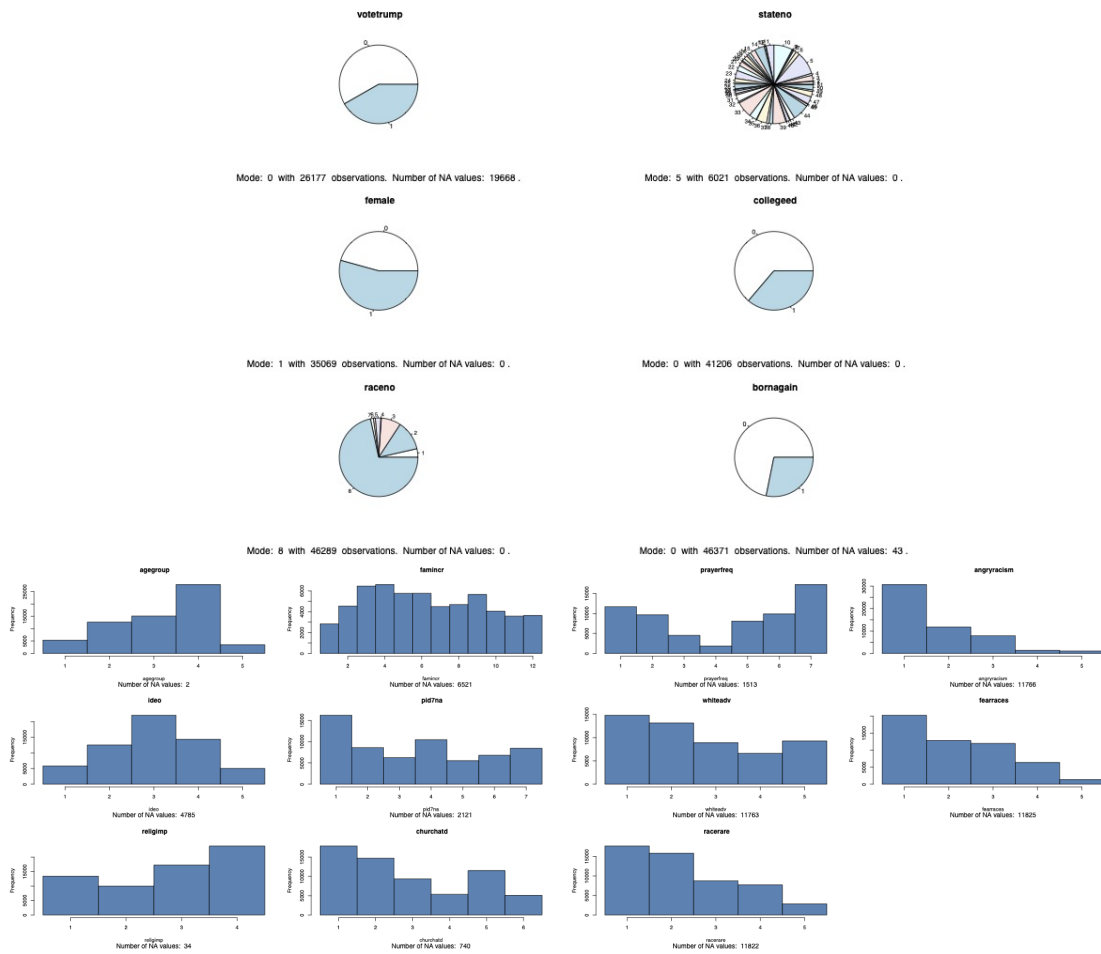


Figure 1: The pie charts illustrate the distributions of the categorical variables. The histograms present the distributions of those variables, where the answer options were on some scale. Each caption informs on the number of missing values corresponding to the specific variable.

Let's take a deeper look at a few variables. For example, the ideology is very symmetrically distributed around its mean value. On the other hand, the distribution of prayer frequency indicates that a US citizen is most likely to either pray a lot or not at all. Moreover, we see that the majority of the respondents aligns with more liberal values in their ideology, racial beliefs, and political beliefs. It is also notable that the age distribution is

very skewed to the left. This may indicate either that the age groups are not well-formed or that people between 50-74 years of age are more prone to vote than younger people. However, the data serves well in this purpose, since it could be assumed that people born within the same twenty-year period possess somewhat similar beliefs and values.

From the pie charts, we notice that the vast majority of the respondents identified themselves as racially white. Moreover, although the majority of respondents did not self-identify as born-again Christians, the number of participants who identified as one exceeds a fourth of the total number of respondents.

2.1 Bivariate analysis

In this section, we conduct a bivariate analysis for a few variables of special interest. In all cases, we firstly form the contingency tables by counting the number of observations representing each distinct pair of values. For example, for the partianship score and whether an individual voted for Trump, we get Table 2.

Voted Trump	Partianship score						
	1	2	3	4	5	6	7
No	11580	4764	4152	3214	705	1028	397
Yes	362	882	238	2844	3679	4033	6442

Table 2: Contingency table of partianship score and Trump vote. Recall that the values range from 1=strong democrat to 7=strong republican.

We further compute the row and column profiles and the corresponding matrix one would assume under if these variables were independent. Then we can form the attraction-repulsion matrix, that is shown in Table 3.

Voted Trump	Partianship score						
	1	2	3	4	5	6	7
No	1.66	1.45	1.62	0.91	0.28	0.35	0.10
Yes	0.07	0.37	0.13	1.13	2.01	1.91	2.26

Table 3: Attraction-repulsion matrix of partianship score and Trump vote. Recall that the values range from 1=strong democrat to 7=strong republican.

Clearly, strong republican tendencies seem to attract Trump voting, since the attraction-repulsion score rises above 2 among the strongest republicans.

Other, more interesting observations related to Trump voting, are presented in the following tables.

Voted Trump	The importance of religion			
	1	2	3	4
No	1.40	1.12	0.94	0.77
Yes	0.44	0.83	1.09	1.33

Table 4: Attraction-repulsion matrix of religious importance and Trump vote. Recall that the values range from 1=not at all religious to 4=very religious.

Voted Trump	Age group				
	1	2	3	4	5
No	1.33	1.24	1.08	0.91	0.73
Yes	0.55	0.66	0.88	1.13	1.37

Table 5: Attraction-repulsion matrix of age groups and Trump vote. Recall that the values range from 1=18-24 years old to 5=75-90 years old.

Voted Trump	Respondent is angry that racism exists				
	1	2	3	4	5
No	1.28	0.79	0.42	0.32	0.32
Yes	0.61	1.30	1.81	1.95	1.957

Table 6: Attraction-repulsion matrix of anger towards racism and Trump vote. Recall that the values range from 1=strongly agrees to 5=strongly disagrees.

Voted Trump	College degree	
	No	Yes
No	0.90	1.15
Yes	1.15	0.79

Table 7: Attraction-repulsion matrix of college education and Trump vote.

Based on the attraction-repulsion matrices in Tables 4-7, people who voted for Trump tend to be more religious than not, above 50 years old, less likely to be angry that racism exists, and less likely to possess a college degree.

Another interesting example that does not include voting for Trump is the correspondence of being angry that racism exists and being religious in Figure 2. The figure indicates that less religious people align much with those who feel angry for racism existing. On the other hand, the arrow indicating the least anger towards racism aligns nearly perfectly with the arrow that indicates major religious importance.

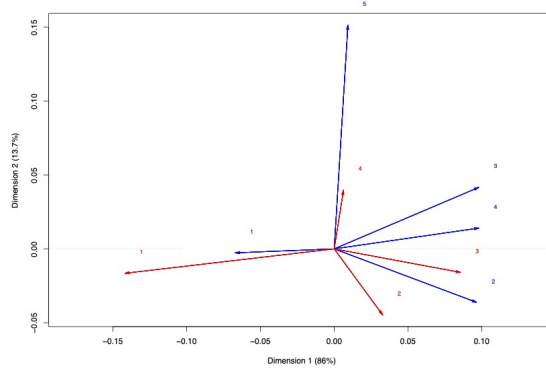


Figure 2: The blue arrows represent the anger towards racism (scale is the same as before) and the red arrows represent the importance of religion.

3 Multiple correspondence analysis

This section explores the relationships between the answers to the 16 questions. This exploration is conducted using multiple correspondence analysis (MCA) which is chosen due to the categorical nature of the data. Indeed, the data contain only answers to the questions that the respondent had to choose one option of the answers. Therefore, the data can be decomposed as an indicator matrix. The indicator matrix here consists of the modalities of 16 variables in Table 1. To this matrix, we apply a similar component analysis approach as in the bivariate case.

As mentioned earlier, the data contains many missing values. The majority of these missing values are associated with the variable "votetrump", i.e. whether the respondent voted for Donald Trump or not. Moreover, the questions related to the respondent's attitudes toward racism ("angryracism", "whiteadv", "ferraces", "racerare") are missing many responses. The reason why these data are missing remains unclear to me. However, if there is a specific reason why these data were missing, removing the observations that lack responses to these questions can bias the results. For example, if the respondent has refused to answer certain questions, this refusal itself may provide valuable information on the attitudes of the respondent. Therefore, we keep all the data points.

As the main interest of this project lies in investigating whether a response to a certain question typically appears with a specific response to another question, we first apply MCA on the column profiles. By doing this, we hope to reduce the dimensionality of the data. Figure ?? illustrates, the variation explained by each MCA component.

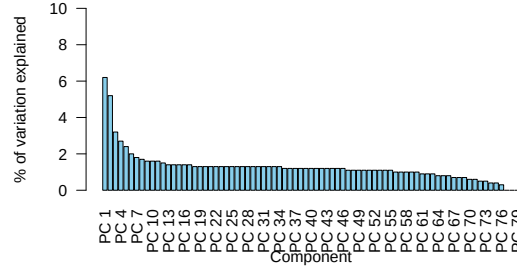


Figure 3: Variation explained by each MCA component.

Figure 3 shows that the first two components explain roughly 11% of the total variation. Although this percentage is not very high, which indicates that there is much variation that is due to other factors than those that vastly contribute to these components, for simplicity, we proceed analysis with the first two components.

Variable	qlt	k = 1		k = 2	
		cor	ctr	cor	ctr
votetrump:0	549	112	14	437	63
votetrump:1	565	200	29	365	63
votetrump:NA	649	637	90	12	2
ideo:1	150	0	0	149	33
ideo:2	161	4	1	157	31
ideo:4	226	20	3	206	39
ideo:5	139	12	2	126	28
pid7na:1	171	6	1	164	30
pid7na:7	218	20	4	198	42
bornagain:0	246	0	0	246	17
bornagain:1	246	0	0	246	43
religimp:1	303	0	0	302	58
religimp:4	254	3	0	252	39
churchatd:1	244	0	0	244	43
prayerfreq:1	256	2	0	254	51
prayerfreq:7	164	9	1	155	28
angryracism:1	348	139	15	209	27

Table 8: Quality of representation (qlt), quality of representation on the corresponding axis (cor), and contribution on the axis (ctr). All the values are $\times 10^{-3}$.

Table 8 presents variables, whose quality of representation was more than 1% on the first MCA axes. Indeed, from the table, we can interpret, that of these variables the first axis is mostly constructed based on the information on whether the individual has answered the voting question. In addition, other voting outcomes have little contribution to this axis. The voting variables also seem to contribute on the construction of the second axis. Moreover, the axis seems to represent religious beliefs.

Table 8 also shows that almost two-thirds of the variation of not answering the voting question is explained by the first axis. This axis also explains a significant proportion of

the variance of voting outcomes and some variance of racial beliefs. The second axis well explains the variation of a much wider group of variables.

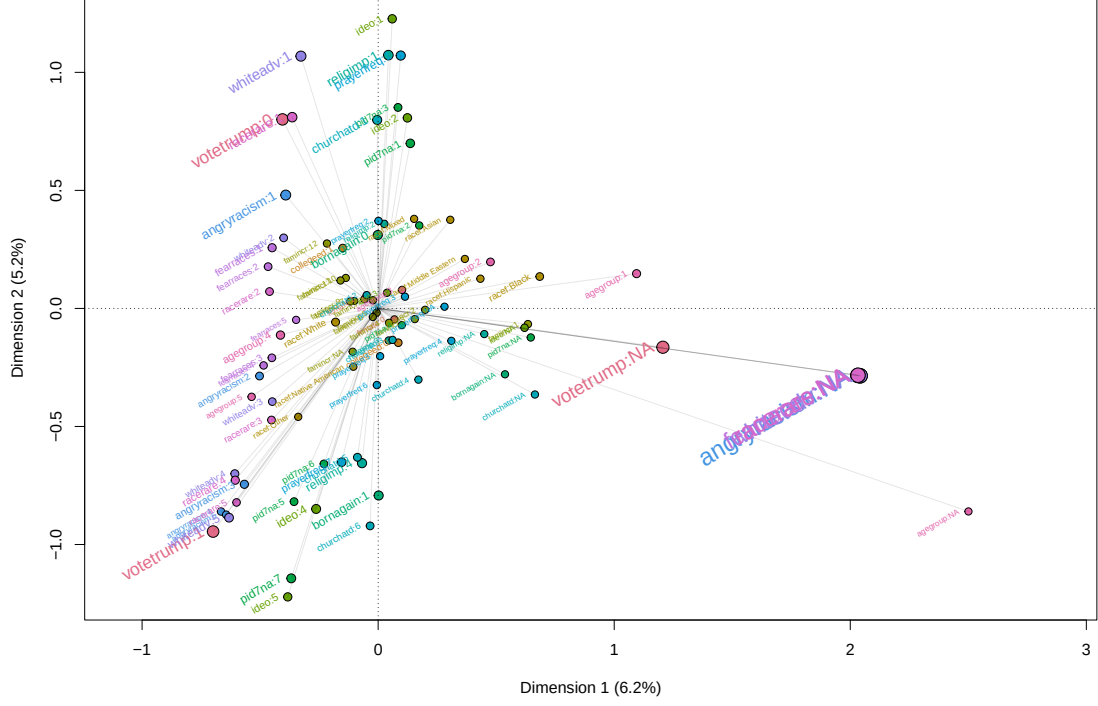


Figure 4: Different variables and their modalities on the first two MCA axes. The sizes of the balls are scaled to represent the quality of representation.

Figure 4 illustrates different variables and their modalities on the first two MCA axes. The x-axis seems to represent, whether a question was answered or not. The y-axis on the other hand aligns well with voting for Trump. In particular, the positive y-axis seems to represent more liberal values, since the points very high on the axis represent e.g. liberal ideology, being very angry for racism existing, and believing that white people have advantages over people from other ethnic backgrounds. The negative y-axis consists of variables indicating devoting to a religion, not believing in racism, and having a conservative ideology.

4 Conclusion

Indeed, since voting for Trump is well-represented by the first two MCA components, we may interpret that the people voting for Trump indeed share at least some of the values on the negative y-axis in Figure 4. Moreover, by Figure 4, the people who did not answer one of the racial questions did not seem to answer any racial questions nor the voting question and were much more likely to have skipped other questions as well. Also, these people may have been on the conservative side of ideologies. One potential answer of the question of why these people have not answered these questions could be that these individuals were not comfortable sharing this information or did not consider such questions important.

Although only a small proportion is explained by the variance of the first two MCA components, the results make sense. Thus, it may be that people who vote for Trump share similar ideologies.

References

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